

MTH203: LECTURE 03

We will first review some more preliminary notions from set theory. After that, we will begin our study of the real numbers.

1. CARDINALITY OF POWER SETS

Recall that for any set S , $\mathcal{P}(S)$ denotes the power set of S , i.e. the set of all subsets of S . It is easy to see that if S has n elements, then $\mathcal{P}(S)$ has 2^n elements. (Exercise: Try to prove this.) Clearly, there does exist an *injective function* from S into $\mathcal{P}(S)$. Indeed, if we define $\theta : S \rightarrow \mathcal{P}(S)$ to be the function $\theta(s) = \{s\}$, it is easy to see that this is an injective function. However, we will see that there is no bijection between S and $\mathcal{P}(S)$.

Theorem 1. *Let S be a set. There is no bijection $f : S \rightarrow \mathcal{P}(S)$.*

Proof. Suppose that there exists a bijection $f : S \rightarrow \mathcal{P}(S)$. Then, let $g : \mathcal{P}(S) \rightarrow S$ be the inverse function of f .

We say that $x \in S$ is *good* if x is not an element of the subset $f(x)$ of S . Let T be the set of all the good elements of S . So, symbolically we may write

$$T = \{x \in S \mid x \notin f(x)\}.$$

Let $y = g(T)$, so that $f(y) = f \circ g(T) = T$. Then, we check whether y is a good or not.

Suppose y is good. Then, by the definition of T , we must have $y \in T$. But then this means that $y \in f(y)$, and hence y cannot be good. This contradicts our assumption that y is good. Thus, this assumption must be false.

Thus, y is not good. This means that $y \in f(y) = T$. But, by definition, an element of S is in T if and only if it is good. Thus, y cannot be an element of T . This is a contradiction.

Thus, we see that our assumption about the existence of the bijection f must be wrong. Thus, there is no bijection between S and $\mathcal{P}(S)$. $\square$

Remark 2. This argument essentially encodes the famous “Epimendes paradox”. Consider the statement, “This statement is false.” If this statement is true, it must be false. However, if it is false, it must be true. This has very interesting implications in logic. See the reading suggestions below.

Thus, we see that for any set S , the set $\mathcal{P}(S)$ is essentially “bigger”. In particular, we see that the set $\mathcal{P}(\mathbb{N})$ is *not countable*. Thus, we see that *uncountable sets exist*.

2. INDUCTION

A very useful method of proof is the method of *induction*. This is based on a very important property of the natural numbers, related to the usual order relation on the natural numbers.

First, let us recall what an order relation is.

Definition 3. Let S be a set. An *order relation* on S is a relation $\leq$ that has the following properties:

- (a) (Reflexivity) $x \leq x$ for all $x \in S$.
- (b) (Antisymmetry) If $x \leq y$ and $y \leq x$ for any $x, y \in S$, then $x = y$.
- (c) (Transitivity) If $x \leq y$ and $y \leq z$ for $x, y, z \in S$, then $x \leq z$.

A set equipped with an order relation is called a *partially ordered set*.

We say that the order relation is a *total order* if it satisfies the following extra condition:

- (d) For any $x, y \in S$, we have either $x \leq y$ or $y \leq x$.

A set equipped with a total order is called a *totally ordered set*.

Notation 4. If S is a partially ordered set, with the order relation being denoted by $\leq$, then we will sometimes write $x \leq y$ as $y \geq x$.

We will assume an understanding of the usual order relation on the natural numbers. This order relation has the following very important property:

Theorem 5 (Well-ordering principle). *Let X be a non-empty subset of $\mathbb{N}$. Then X has a least element. (In other words, there exists an element $x_0 \in X$ such that $x_0 \leq x$ for any $x \in X$.)*

We will not look at the proof of this statement since we have not really looked at the construction of the natural numbers either. We will simply assume this property of the integers. This theorem gives us the principle of *induction*.

Let $\mathcal{S}(n)$ be a statement about the integer n and suppose that we are trying to prove that $\mathcal{S}(n)$ is true for all integers n . This can be done by the following steps:

- **Step 1:** Prove that $\mathcal{S}(1)$ is true. (In practice, this may turn out to be a very simple calculation.)
- **Step 2:** Assume that $\mathcal{S}(k)$ is true for some k and deduce from this that $\mathcal{S}(k+1)$ is true.

Then, it follows that $\mathcal{S}(n)$ is true for all natural numbers n . Indeed, suppose this is not true. Let X be the set of all natural numbers n such that $\mathcal{S}(n)$ is not true. Then, the set X is *non-empty*. By the Well-ordering principle, X has a least element, which we denote by n_0 . Then $n_0 - 1$ does not lie in X and so $\mathcal{S}(n_0 - 1)$ is true. Then, by Step 2, $\mathcal{S}(n_0 - 1 + 1)$ is true, i.e. $\mathcal{S}(n_0)$ is true. But then n_0 cannot be an element of X . This contradiction shows our assumption is false, i.e. $\mathcal{S}(n)$ is true for all natural numbers n .

Example 6. Let us show that $\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$. We will prove this by induction on n .

First we verify that the statement holds for $n = 1$. This is easy since

$$\sum_{i=1}^1 i^2 = 1^2 = 1 = \frac{1 \times (1+1) \times (2 \times 1 + 1)}{6}.$$

Now, suppose that we have $\sum_{i=1}^k i^2 = \frac{k(k+1)(2k+1)}{6}$. Then,

$$\begin{aligned}
 \sum_{i=1}^{k+1} i^2 &= \left(\sum_{i=1}^k i^2 \right) + (k+1)^2 \\
 &= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 && \text{(by assumption)} \\
 &= (k+1) \left(\frac{k(2k+1) + 6(k+1)}{6} \right) \\
 &= (k+1) \left(\frac{2k^2 + k + 6k + 6}{6} \right) \\
 &= (k+1) \left(\frac{2k^2 + 4k + 3k + 6}{6} \right) \\
 &= (k+1) \left(\frac{2k(k+2) + 3(k+2)}{6} \right) \\
 &= \frac{(k+1)(k+2)(2(k+1)+1)}{6}.
 \end{aligned}$$

Thus, the statement holds for the integer $k+1$. Thus, by the principle of induction, it is true for all natural numbers.

Sometimes, Step 2 in the above description is modified as follows:

- **Step 2:** Let k be a natural number and assume that $\mathcal{S}(l)$ is true for all $l \leq k$. Deduce from this that $\mathcal{S}(k+1)$ is true.

It is easy to verify (using essentially the same argument as above) that this variation also allows us to prove $\mathcal{S}(n)$ for all natural numbers n .

3. FIELDS

We would now like to turn our attention to the main topic of this course – the real numbers. We will not construct the real numbers, but we will describe their fundamental properties. For this, we first look at the notion of a *field*.

A *field* is a set F along with two functions $+$ (called as *addition*) and $\cdot$ (called as *multiplication*) from $F \times F$ to F . For these functions, we use a slightly different notation. The image of (a, b) under the function $+$ is written as $a + b$ (and not $+(a, b)$) and its image under the function $\cdot$ is written as $a \cdot b$, or just ab . These functions are required to satisfy some properties, which we will now list below.

Properties of addition:

- (A1) (Associativity) $a + (b + c) = (a + b) + c$ for all $a, b, c \in F$.
- (A2) (Identity) There exists an element $0 \in F$ such that $0 + a = a + 0 = a$ for all $a \in F$.
- (A3) (Inverses) For any element $a \in F$, there exists an element $-a$ such that $a + (-a) = (-a) + a = 0$.
- (A4) (Commutativity) $a + b = b + a$ for all $a, b \in F$.

Properties of multiplication:

- (M1) (Associativity) $a \cdot (b \cdot c) = (a \cdot b) \cdot c$ for all $a, b, c \in F$.
- (M2) (Identity) There exists an element $1 \in F$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in F$.
- (M3) (Inverses) For any element $a \in F$, such that $a \neq 0$, there exists an element a^{-1} such that $a \cdot a^{-1} = a^{-1} \cdot a = 1$.
- (M4) (Commutativity) $a \cdot b = b \cdot a$ for all $a, b \in F$.

Distributivity property:

$$(D) \quad a \cdot (b + c) = a \cdot b + a \cdot c \text{ for all } a, b, c \in F.$$

Example 7. The rational numbers (denoted by $\mathbb{Q}$), the real numbers (denoted by $\mathbb{R}$) and the complex numbers (denoted by $\mathbb{C}$) are some common examples of fields. There are many other fields which are of interest in mathematics, but we will not need to look at them in this course.

Example 8. The natural numbers (denoted by $\mathbb{N}$) and the integers (denoted by $\mathbb{Z}$) are also equipped with operations of multiplication and addition. These will satisfy most of the properties listed above. However, $\mathbb{N}$ fails to satisfy (A3) and (M3), while $\mathbb{Z}$ fails to satisfy (M3). Thus, these are not fields.

In this course, we are going to assume that we understand what $\mathbb{Q}$ is. (We will describe certain basic properties that $\mathbb{R}$ satisfies in relation to $\mathbb{Q}$.) Apart from the above structure, $\mathbb{Q}$ has yet another property – it is a totally ordered set. In fact, it is an “ordered field”, which we will define below.

4. ORDERED FIELDS

An *ordered field* is a field F , equipped with a total order $\leq$ such that the following two properties hold:

(O1) If $x \leq y$, then $x + z \leq y + z$ for any $z \in F$.

(O2) If $x \geq 0$ and $y \geq 0$, then $xy \geq 0$.

These two properties have some implications which should be familiar. We prove one consequence as an example:

Lemma 9. *Let F be an ordered field. Let x and y be elements of F such that $x \leq y$. Then $xz \leq yz$ for any $z \geq 0$.*

Proof. By (O1) (choosing z to be equal to $-x$, we get $0 = x + (-x) \leq y + (-x) = 0$. Thus, $y + (-x) \geq 0$.

If $z \geq 0$, by (O2), see that $(y + (-x))z \geq 0$. Thus, $yz + (-x)z \geq 0$. By (O1), (adding xz to both sides), we see that $yz + (-x)z + xz \geq 0 + xz$. Thus,

$$yz + [(-x) + x]z \geq xz$$

and hence $yz \geq xz$. □

5. LEAST UPPER BOUNDS

The field $\mathbb{Q}$ is an ordered field, where the order relation is the usual one. If x, y, z, w are integers, with y and w being non-zero, then $x/y \geq z/w$ if and only if $xw - yz$ is in $\mathbb{N} \cup \{0\}$.

We would like $\mathbb{R}$ to be a field containing $\mathbb{Q}$. We would like it to be an ordered field such that its order relation is an extension of the order relation on $\mathbb{Q}$. (This means that if x and y are rational numbers, then $x \leq y$ when viewed as real numbers if and only if $x \leq y$ when viewed as rational numbers.) It might be possible to construct many ordered fields like this, but we will impose an additional condition that will uniquely characterize the real numbers. This will be the *axiom of completeness*. Before we state what this is, we require a little terminology.

Given a partially ordered set S , with order relation $\leq$, and a subset T of S , an element $s \in S$ is said to be an *upper bound of T* if $s \geq t$ for all $t \in T$. We say that a subset $T \subset S$ is *bounded above* if it has an upper bound in S . Analogously, we can define the notion of a *lower bound* of a set and the corresponding notion of sets being *bounded below*.

Given a subset T of a partially ordered set S , a *least upper bound* of T is an upper bound s_0 of T such that if s is any other upper bound of T , then $s_0 \leq s$. (Analogously, we can define the notion of a *greatest lower bound* of a set.)

Definition 10. Let S be a partially ordered set. We say that S has the *least upper bound property (LUB property)* if every *non-empty*, bounded above subset of S has a least upper bound.

Let us first observe that $\mathbb{Q}$ does not have this property.

Example 11. Let us consider $\mathbb{Q}$ with the usual order relation. Let T be the subset of $\mathbb{Q}$ defined by

$$T = \{x \in \mathbb{Q} \mid x^2 < 2\}.$$

We claim that this set is bounded above.

Lemma 12. Let α be any rational number such that $\alpha^2 > 2$ and $\alpha > 0$. Then, α is an upper bound for T .

Proof. Suppose that α is not an upper bound of T . Then there exists some $x \in T$ such that $x > \alpha$. Thus, $x - \alpha > 0$. However, $x^2 < 2 < \alpha^2$. Thus, $0 > x^2 - \alpha^2 = (x - \alpha)(x + \alpha)$. As $x - \alpha > 0$, we must have $x + \alpha < 0$ (since the product of the two numbers, $x - \alpha$ and $x + \alpha$ is negative). Thus, $x - \alpha > x + \alpha$, which implies that $2\alpha < 0$. This contradicts the assumption that $\alpha > 0$. This shows that α must be an upper bound for T . $\square$

We now claim that T does not have a least upper bound in $\mathbb{Q}$. Indeed, suppose β is a least upper bound of T . Note that $0 \in T$ and so $\beta > 0$.

First, let us show that $\beta^2 > 2$. Suppose $\beta^2 < 2$. Then, we will construct an element $\gamma \in T$ such that $\gamma > \beta$, which will contradict the fact that β is an upper bound of T . We will choose γ to be of the form $\beta + (1/n)$ for an appropriately chosen natural number n . We want n to be such that $\gamma^2 < 2$. In other words,

$$\beta^2 + 2\beta/n + 1/n^2 < 2,$$

or equivalently,

$$\frac{2\beta + 1/n}{n} < 2 - \beta^2.$$

For this, we simply choose n to be any integer greater than $(2\beta + 1)/(2 - \beta^2)$. (Note that $(2\beta + 1)/(2 - \beta^2) > 0$ since $\beta > 0$ and $2 > \beta^2$.) Indeed, once n has been chosen in this manner, we have

$$\frac{2\beta + 1/n}{n} \leq \frac{2\beta + 1}{n} < 2 - \beta^2$$

as required. Then, setting $\gamma = \beta + 1/n$, we see that $\gamma^2 < 2$. This gives us the required contradiction and we conclude that $\beta^2 > 2$.

Then, we claim that there exists some β_1 such that $0 < \beta_1 < \beta$ and $\beta_1^2 > 2$. This can be done by a similar argument to the one above. We seek to find some integer n such that β_1 can be taken to be $\beta - 1/n$. Thus, we require the inequality

$$\beta^2 - 2\beta/n + 1/n^2 > 2,$$

or equivalently

$$\frac{\beta^2 - 2\beta + 1/n}{n} < \beta^2 - 2.$$

We choose n to be any natural number larger than $2\beta/(\beta^2 - 2)$. (Note that $2\beta/(\beta^2 - 2) > 0$.) Then we have

$$\frac{2\beta - 1/n}{n} < \frac{2\beta}{n} < \beta^2 - 2$$

as required. Then, if we set $\beta_1 = \beta - 1/n$, we get $\beta_1^2 > 2$.

Since $(6/5)^2 < 2$, we have $6/5 \in T$ and so $\beta > 6/5$. Thus, $\beta_1 = \beta - 1/n \geq 6/5 - 1 > 0$. Thus, we have constructed β_1 such that $0 < \beta_1 < \beta$ and $\beta_1^2 > 2$.

By Lemma 12, we see that β_1 is an upper bound for T . Since $\beta_1 < \beta$, we have a contradiction.

6. READING SUGGESTIONS

This lecture corresponds to Sections 1.2, 1.3 and (some part of) 1.6 in the text ([1]).

The following reading suggestions have nothing to do with this course, but are loosely connected to what was discussed in this lecture:

- The Epimenes paradox has very interesting implications in logic. For a very interesting account of this, along with an exploration of related themes in music and other arts, see the popular book [3] by Douglas Hofstadter.
- Look up the *Continuum Hypothesis* on the internet (or read Section 25 from [2]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. 6
- [2] Paul R. Halmos, *Naive Set Theory*, Undergraduate Texts in Mathematics, Springer, 1974. 6
- [3] Douglas Hofstadter, *Gödel, Escher, Bach: an Eternal Golden Braid*,